Primary Years Programme

Subject continuums: Science

International Baccalaureate®
Baccalauréat International
Bachillerato Internacional

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The International Baccalaureate Organization (known as the IB) offers four high-quality and challenging educational programmes for a worldwide community of schools, aiming to create a better, more peaceful world. This publication is one of a range of materials produced to support these programmes.

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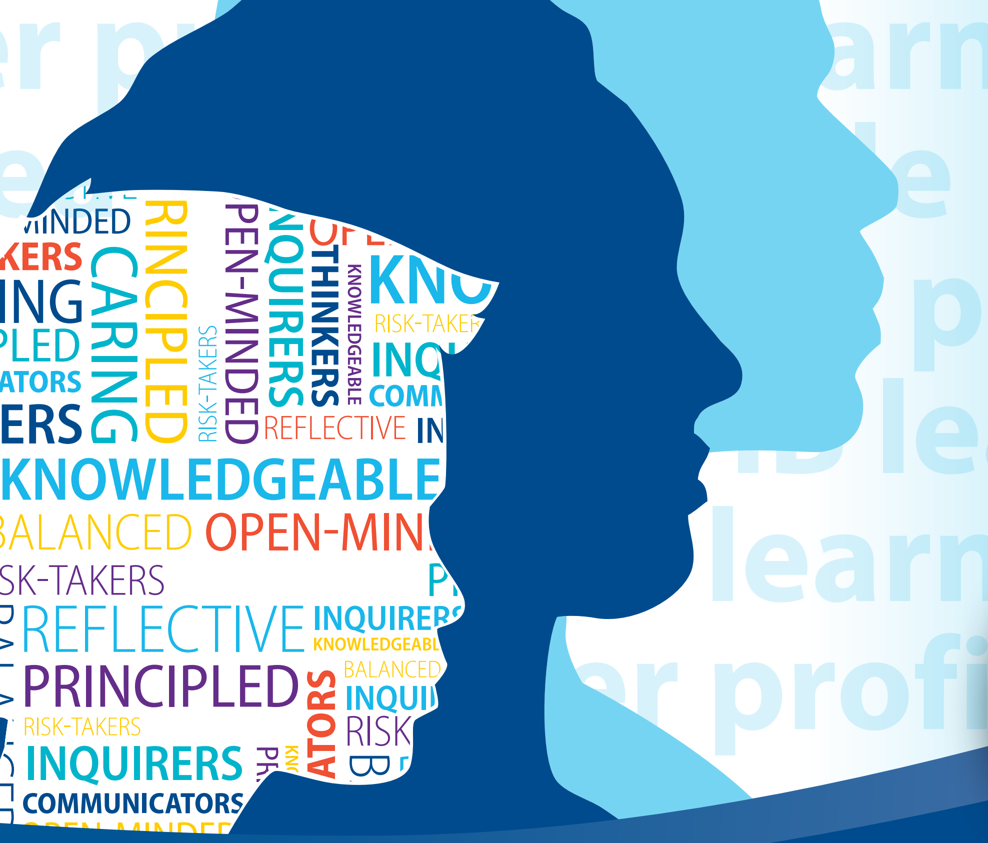
IB mission statement

The International Baccalaureate aims to develop inquiring, knowledgeable and caring young people who help to create a better and more peaceful world through intercultural understanding and respect.

To this end the organization works with schools, governments and international organizations to develop challenging programmes of international education and rigorous assessment.

These programmes encourage students across the world to become active, compassionate and lifelong learners who understand that other people, with their differences, can also be right.



THE IB LEARNER PROFILE

IB learner profile

The aim of all IB programmes is to develop internationally minded people who, recognizing their common humanity and shared guardianship of the planet, help to create a better and more peaceful world.

As IB learners we strive to be:

INQUIRERS
We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.

KNOWLEDGEABLE
We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.

THINKERS
We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions.

COMMUNICATORS
We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.

PRINCIPLED
We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.

OPEN-MINDED
We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.

CARING
We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.

RISK-TAKERS
We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.

BALANCED
We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.

REFLECTIVE
We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.

The IB learner profile represents 10 attributes valued by IB World Schools. We believe these attributes, and others like them, can help individuals and groups become responsible members of local, national and global communities.

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Contents

Science strands

Living things	An exploration of the *sustainability*, complexity and interconnectedness of life on our planet. Through scientific and traditional knowledges, learners explore the natural world, the patterns and characteristics that defines it, and the ways in which living things change and grow over time; how living things are organized, each with its own unique *phenomena* and behaviours, and how living things adapt to changes in their *environment* over time; the ways in which we rely on each other to thrive and survive. Learners investigate the relationships and interdependencies that exist between different species including ourselves; a deeper understanding of the living things that surround us, helping us to become informed and responsible stewards of our planet.
Earth and space	An exploration of the structure of our planet and its position in the solar system; an understanding of the properties, characteristics and changes of our planet, including diverse ecosystems, natural cycles and the use of resources that shape our planet. Learners inquire into the interconnectedness of Earth's systems and how natural phenomena and human activity impact Earth, and how humans and other living things have adapted to changes in the environment. Learners reflect on the importance of sustainability to maintain a healthy planet for future generations; how humans observe and gather data to appreciate where our understanding has come from, and how models help explain the development of the universe over time.
Physics and chemical science	An exploration of physical objects and chemical substances, matter and materials, as well as the laws governing their behaviour; properties of matter as well as processes that change these properties, and how we can manipulate them to create new materials or products; how people can explain and predict the behaviours of various substances, and help us develop new materials with specific properties; the ways in which people design technologies and applications, and reflect on the potential impact of these on society and the environment.

Overall expectations

Overall expectations			
All strands			
Phase 1	Phase 2	Phase 3	Phase 4
Learners develop their observational skills by using each of their senses to gather and record information, identify patterns, ask questions/make guesses and discuss their ideas. Learners explore different tools through play to discover the way objects and phenomena function, and recognize basic cause-and-effect relationships. Learners are aware of different perspectives and examine change over a variety of time periods. Learners identify the relationship between themselves and the environment around them, showing care and respect for themselves, other living things and the environment. Learners communicate their ideas and provide explanations using their own scientific experience and vocabulary.	Learners increase their observational skills by using their senses to gather and record information in order to identify patterns, make predictions and refine their ideas. Learners select different tools and explore ways to use them for their investigations/inquiry. Learners identify a question or problem to be explored. Learners discover the way objects and phenomena function, identify parts of a system and gain an understanding of cause-and-effect relationships. Learners examine varying time periods and develop their understanding that variables and conditions may affect change. Learners are aware of different perspectives and the *interdependence* between the environment and us, recognizing the need to respect each other, living things and the environment. Learners communicate their ideas and provide explanations using their own scientific experience and understanding.	Learners use their observational skills by using their senses and selecting observational tools to gather and record observed information in a number of ways. They reflect on these findings to identify patterns or connections, make predictions/*hypotheses*, and test and refine their ideas with increasing accuracy. Learners explore the way objects and phenomena function, identify parts of a system and gain an understanding of increasingly complex cause-and-effect relationships. They examine change over time and recognize that change may be affected by one or more variables. Learners take into consideration different perspectives and how these views may have been formulated. They examine how products and tools have been developed through the application of scientific investigation. Learners consider ethical issues in science-related contexts and use their learning to plan thoughtful and realistic action, in order to improve their welfare, that of other living things and the environment. Learners communicate their ideas and provide explanations using their own scientific experience and that of others.	Learners use their observational skills by selecting observational tools to gather and record observed information in a number of ways. They reflect on these findings to identify patterns or connections, make predictions, and test and refine their ideas with increasing accuracy. Learners analyse the functioning of objects and phenomena, identify components within systems and develop an understanding of progressively intricate cause-and-effect connections. They observe changes over time and acknowledge that these changes can be influenced by one or more variables. Learners reflect on the impact that the application of science, including advances in *technology*, has had on us, society and the environment. Learners examine different perspectives and how these views may have been formulated. They analyse ethical and social issues in science-related contexts and express their responses. Learners use their acquired knowledge to effectively communicate their ideas and strategize thoughtful and practical actions aimed at enhancing their own and others' well-being and the environment.

Strand: Living things

Example concepts: adaptation, animals, biodiversity, biology, change, classification, *conservation*, ecosystems, environment, ethics, evolution, genetics, growth, habitat, health, homeostasis, interactions, organism, pattern, plants, relationships, survival, sustainability, systems, technology.

<table>
<tr><th colspan="4">Example conceptual understandings</th></tr>
<tr><th colspan="4">Strand: Living things</th></tr>
<tr><th>Phase 1</th><th>Phase 2</th><th>Phase 3</th><th>Phase 4</th></tr>
<tr><td>Understanding patterns helps people learn about the world at all levels.</td><td>Patterns help people observe phenomena and make predictions.</td><td>Patterns can be used to make predictions, support cause-and-effect relations and as evidence to support ideas.</td><td>Mathematical representations and empirical evidence are needed to identify patterns.</td></tr>
<tr><td>Living things go through processes of change.</td><td>Over time, living things need to adapt to survive.</td><td>Changes, sudden or gradual, have an effect on stability and sustainability.</td><td>Mathematical knowledge, understanding and modelling help to identify and predict change over time.</td></tr>
<tr><td rowspan="2">Living things have different characteristics to help identify and classify them.</td><td>There are connections between the characteristics of living things and their environment.</td><td>Understanding systems in nature helps recognize how they work and how parts influence and interact with one another to perform a function.</td><td>Humans observe and use different features of the natural world in the designed world.</td></tr>
<tr><td colspan="2">Living things and non-living things are often organized into different levels (organelle, cells, tissues, organs, organ systems, organisms, populations, communities, ecosystem, biosphere).</td><td>Exploring human actions on physical and natural environments helps understand environmental consequences.</td></tr>
<tr><td>Identifying and observing local environments help in the understanding of how to care for them.</td><td>The interdependence of living beings and their environments affects living beings.</td><td>Interactions between and within systems enhance the understanding of connections in environments.</td><td>Living systems are open, self-organizing life forms that interact with their environment.</td></tr>
</table>

Example learning outcomes			
Strand: Living things			
Phase 1	**Phase 2**	**Phase 3**	**Phase 4**
Learners: • identify, sort and classify living things and non-living things • identify that living things have basic needs, in order to stay healthy and survive • plan and carry out or participate in an investigation into how living things survive • model the differing movements of living things (for example, how rabbits hop, how elephants use their trunks, how dolphins swim) • describe how animals are a life-sustaining resource for us and other living beings • describe how plants are a life-sustaining resource for us and other living beings.	Learners: • identify and describe examples of the external features and basic needs of living things • describe how different places meet the needs of living things • discuss how the environment affects humans and other living things • analyse how investigations can help understand how living things survive • identify and describe the changes in living things.	Learners: • compare key stages in the life cycle of a plant and an animal, and relate them to growth and survival • design an investigation to determine what a living thing needs to grow/survive • make observations to construct an evidence-based account to show that young plants and animals are like, but not exactly like, their parents • communicate understandings supported by evidence for how living things can adapt their environment to meet their needs • predict the effects when living things are removed or die out (food webs, food chains).	Learners: • make observations of plants and animals to compare the diversity of life in different habitats • use materials to design a solution to a problem by mimicking how plants and/or animals use their features • design a model that mimics the function of an animal in dispersing seeds or pollinating plants • identify how the reproduction of living things connects to the interdependency of species and their survival, as well as maintaining the environment.

Strand: Earth and space

Example concepts: atmosphere, climate, conservation, cycles, environment, erosion, evidence, geography, geology, gravity, motion, pattern, *renewable* and non-renewable energy sources, resources, seasons, space, sustainability, systems, tectonic plate movement.

Example conceptual understandings			
Strand: Earth and space			
Phase 1	**Phase 2**	**Phase 3**	**Phase 4**
Systems have parts that work together.	We live in a world of interacting systems in which the actions of any individual element affect others.	A system is a group of related parts that make up a whole which can carry out functions its individual parts cannot.	Models can be used to simulate systems and interactions between them, and to predict the behaviour of a system.
Earth as part of a system orbits around the Sun.	Earth's natural cycles and human activity are usually connected.	Natural phenomena can be observable on Earth due to movements and rotations.	Sudden geological changes and extreme weather conditions affect the Earth's surface.
Observable patterns and cycles occur in the local environment.	Earth's surface changes, over time, as a result of natural processes and human activity.	Some of Earth's resources cycle through the environment, but others are finite.	Human activities in the environment, such as recycling and animal conservation or exploitation of natural resources, have an impact on Earth.
Sustainability requires us to care for Earth's/our resources.	Exploration leads to discovery and develops new understandings.	The development of technology is both a help and a hindrance for sustainability.	Past systems and technologies help shape (new, present, future) *innovations*.
	Earth's resources are used in a variety of ways.	People establish diverse practices to conserve, protect, sustain and maintain the Earth's resources.	Resources must be used sustainably for the benefit of present generations without compromising the needs of future generations.

Example learning outcomes			
Strand: Earth and space			
Phase 1	**Phase 2**	**Phase 3**	**Phase 4**
Learners: • make observations at different times of the day/week/year to relate the amount of daylight to the time of year • identify different aspects of patterns and cycles in their authentic environments (for example, weather, night and day, seasons) • make connections between cycles and changes in the environment and other cultural perspectives • recognize different resources in our environment and how humans use them • identify that the choices we make can have an impact on the land, water, air and other living things • consider different ways to take care of our resources and how we can reduce, reuse and recycle.	Learners: • observe, describe and predict patterns of the motion of the Sun, Moon and stars in the sky • identify the appearance of the Sun, Moon and stars (sunrise/set, moonrise/set) • discuss how the Sun and the Moon are important in different cultures, with respect to customs and traditions • explain how wind and water change the Earth • identify events that occur quickly or slowly on Earth (for example, volcanic activity, earthquakes, erosion) • identify water as a resource that cycles through the environment.	Learners: • understand that Earth rotates on its axis while revolving around the Sun and that the Moon revolves around the Earth • use observations of the Sun, Moon and stars to describe patterns that can be predicted • explore how science has contributed to understanding and resolving issues related to the impact of human activities • compare multiple solutions designed to slow or prevent wind or water from changing the shape of the land.	Learners: • model how the Earth is shaped by water, erosion and other factors • use evidence to analyse and conclude that the Earth's natural processes have patterns • explain why different regions of Earth experience different seasonal conditions • consider what is meant by the term "renewable" in relation to the Earth's resources • identify timescales for *regeneration* of resources • design models to represent systems and their interactions • explore different technologies found in different civilizations • describe the impact of systems and technologies from past civilizations in present times.

Strand: Physical and chemical science

Example concepts: changes of state, chemical and physical changes, classification, conduction and convection, conservation, density, efficiency, environment, equilibrium, forces of energy, forms of energy (electricity, heat, kinetic, light, potential, sound), gases, invention, light, liquids, magnetism, mechanics, movement, physics, pollution, properties and uses of materials, solids, structures, sound, sustainability, technology, transformation of energy.

Example conceptual understandings			
Strand: Physical and chemical science			
Phase 1	**Phase 2**	**Phase 3**	**Phase 4**
Objects and materials have different names and properties.	Objects and materials can have reversible or irreversible changes through physical and chemical processes.	Solids, liquids and gases behave in different ways and have observable properties that help to classify them.	A change of state between solid, liquid and gas can be caused by adding or removing heat.
Materials can be manipulated, mixed and changed.	Motion is a change in position over time.	Forces can impact on how an object moves or changes shape.	The total amount of energy and matter in *closed systems* is conserved.
Objects can be sorted into groups based on their properties.	Matter has mass and takes up space.	All matter is made of atoms which are composed of protons, neutrons and electrons.	Matter is conserved because atoms are conserved in physical and chemical processes.
The way objects move depends on their size and shape.		There are connections between motion and the behaviour of objects and systems.	Changes in a system can be described in terms of energy and matter.
Objects can move in different ways.	Energy may take different forms (for example, energy in fields, thermal energy, energy of motion).	Energy can be transferred in various ways and between objects.	Energy cannot be created or destroyed—it only moves between one place and another place, between objects and/or fields, or between systems.
Light and sound can be generated by different sources.	Different types of vibrations generate sound.	Properties of light and sound depend on their source and the objects with which they interact.	Sound travels through waves.
			Light from a source can be absorbed, reflected and refracted.

Example learning outcomes			
Strand: Physical and chemical science			
Phase 1	**Phase 2**	**Phase 3**	**Phase 4**
Learners: • identify common materials and their use in daily life • identify the properties of materials • gather, compare, sort, classify, order, interpret and describe observable characteristics and properties • observe patterns and draw conclusions • identify different sources of light and sound • explore different ways to produce sound using familiar objects and actions, for example, striking, blowing, scraping and shaking.	Learners: • identify how light and sound are produced by a range of sources • recognize different properties of light and sound through their senses • describe matter as everything that has mass and occupies volume • investigate how liquids and solids respond to changes in temperature, for example, water changing to ice, or melting chocolate • recognize how a complete circuit allows for the flow of electricity • investigate how different energy sources generate electricity • investigate different electrical conductors and insulators • explore how different sounds may be made by making a variety of materials vibrate.	Learners: • illustrate how light and sound can be absorbed, reflected and refracted • describe the properties of air and their practical applications • recognize how heat can be produced in many ways • identify patterns of movement from one object to another • explain how forces can be exerted by one object on another through direct contact or from a distance • describe how the transfer of energy can be tracked as energy flows through a system • explore how changes from solid to liquid and liquid to solid can help us recycle materials • describe and model the structure of the atom: nucleus, protons, neutrons and electrons • compare the mass and charge of protons, neutrons and electrons.	Learners: • identify that in nuclear processes, atoms are not conserved, but the total number of protons plus neutrons is conserved • describe how energy drives the cycling of matter within and between systems • explain how a change in the temperature of an object is related to the gain or loss of heat by the object.

Science glossary

Closed systems	A system where only energy can be transferred/exchanged between the system and its surroundings.
Conservation	A careful preservation and protection of something, for example, the planned management of a natural resource to prevent exploitation, destruction, or neglect.
Environment	The surroundings or conditions in which a person, animal, or plant lives or operates.
Hypothesis	A proposed explanation, prediction or educated guess that needs to be tested through the scientific method.
Innovation	The practical implementation of ideas that results in the introduction of something new.
Interdependence	The dependence of two or more people or things on each other.
Phenomena	An observable event: natural phenomena occur without human input, for example, gravity, tides, volcanoes, sandstorms.
Regeneration	The natural process of replacing or restoring something that is damaged.
Renewable	Capable of being replaced by natural ecological cycles or sound management procedures.
Sustainability	Fulfilling the needs of current generations without compromising the needs of future generations, while ensuring a balance between economic growth, environmental care and social well-being.
Technology	The application of scientific knowledge for practical purposes.